OpenGonio wiki

Explanations, bibliography (APA 7) and background appendices

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OpenGonio wiki

Open-source four-axis gonioreflectometer and 3D scanner. It measures the bidirectional reflectance distribution function (BRDF) of flat samples and scans objects in 3D. Design follows Ansari-Asl et al. (2024), rebuilt for under $630 in parts, with no soldering, drilling, sawing or crimping.

New here? The **project map** on the website (`docs/explore/`) shows the whole project as a clickable map with a guided tour in plain words. Inside (`docs/inside.html`) opens up the software, tests, firmware, wiring and roadmap, and the control app (`docs/app/`) runs in the browser on the simulated machine.

Page	Contents
How it works	BRDF, geometry, the four axes, one measurement
Machine · Hardware	axes, subsystems, cost
Tools and joins · CAD	joins without tools; 3D files, Onshape
Calibration	angles, camera response, absolute reflectance, drift
3D scanning	fixed camera, visual hull, photogrammetry, photometric stereo
Software · Commands	`opengonio`, simulator, every command
Build guide · Theory	how steps work; all equations
Test materials · Extend · Repository map	samples; changing the design; where files are
Paper	manuscript for *HardwareX*, build, status
Bibliography	every scholarly source, APA 7
Appendices A · B · C · D · E	background papers by topic, with what each is used for

Build guide, bill of materials and commands: README. Step checklists: phases.

References

Ansari-Asl, M., Barbieri, M., Obein, G., & Hardeberg, J. Y. (2024). Optical and electromechanical design and implementation of an advanced multispectral device to capture material appearance. *Journal of Imaging*, *10*(3), 55. https://doi.org/10.3390/jimaging10030055

How it works

What is measured

The BRDF f_r is reflected radiance per unit incident irradiance, for one incoming and one outgoing direction (Hanrahan, 2016; Nicodemus et al., 1977):

$$f_r(\omega_i \to \omega_r) = \frac{dL_r(\omega_i \to \omega_r)}{dE_i} \; [\mathrm{sr}^{-1}]$$

Isotropic materials need 3 angles; anisotropic ones (brushed metal, satin) need 4 (Ward, 1992).

Geometry

One zero point; every axis turns about it.

Axis	Moves	Range
M4	light arm around the sample	20–324°
M3	table (sample yaw)	±180°
M2	sample tilt	±90°
M1	sample spin	±180°

- Measuring camera fixed at 350 mm; light 300 mm from the zero point.
- Light on a horizontal ring: M3 and M2 already turn the sample to every direction an overhead light could reach, so the arm stays low, stiff and out of the light path (Ansari-Asl et al., 2024).
- In-plane, the mirror peak sits at M3 = M4 / 2.
- Light within 20° of the camera is out of reach: pure back-scatter is not measured.

One measurement

1. Move to a pose (inverse kinematics from $\theta_i, \theta_r, \Delta\varphi$).
2. Capture at fixed exposure; several exposures for high dynamic range (HDR) (Debevec & Malik, 1997).
3. Subtract dark frame, linearise, divide by the white reference (Calibration).
4. Fit Lambert + GGX microfacet model (Trowbridge & Reitz, 1975; Walter et al., 2007) → game-engine material.

Full equations: Theory.

References

Ansari-Asl, M., Barbieri, M., Obein, G., & Hardeberg, J. Y. (2024). Optical and electromechanical design and implementation of an advanced multispectral device to capture material appearance. *Journal of Imaging*, *10*(3), 55. https://doi.org/10.3390/jimaging10030055

Debevec, P., & Malik, J. (1997). Recovering high dynamic range radiance maps from photographs. *Proc. SIGGRAPH '97*. https://pauldebevec.com/Research/HDR/debevec-siggraph97.pdf

Hanrahan, P. (2016). *Reflection models I: Ideal materials*.

Nicodemus, F. E., Richmond, J. C., Hsia, J. J., Ginsberg, I. W., & Limperis, T. (1977). Geometrical considerations and nomenclature for reflectance. *NBS Monograph*, *160*. https://doi.org/10.6028/NBS.MONO.160

Trowbridge, T. S., & Reitz, K. P. (1975). Average irregularity representation of a rough surface for ray reflection. *Journal of the Optical Society of America*, *65*(5), 531–536. https://doi.org/10.1364/JOSA.65.000531

Walter, B., Marschner, S., Li, H., & Torrance, K. (2007). Microfacet models for refraction through rough surfaces. *Eurographics Symposium on Rendering*.

Ward, G. J. (1992). Measuring and modeling anisotropic reflection. *ACM SIGGRAPH Computer Graphics*, *26*(2), 265–272. https://doi.org/10.1145/142920.134078

The machine

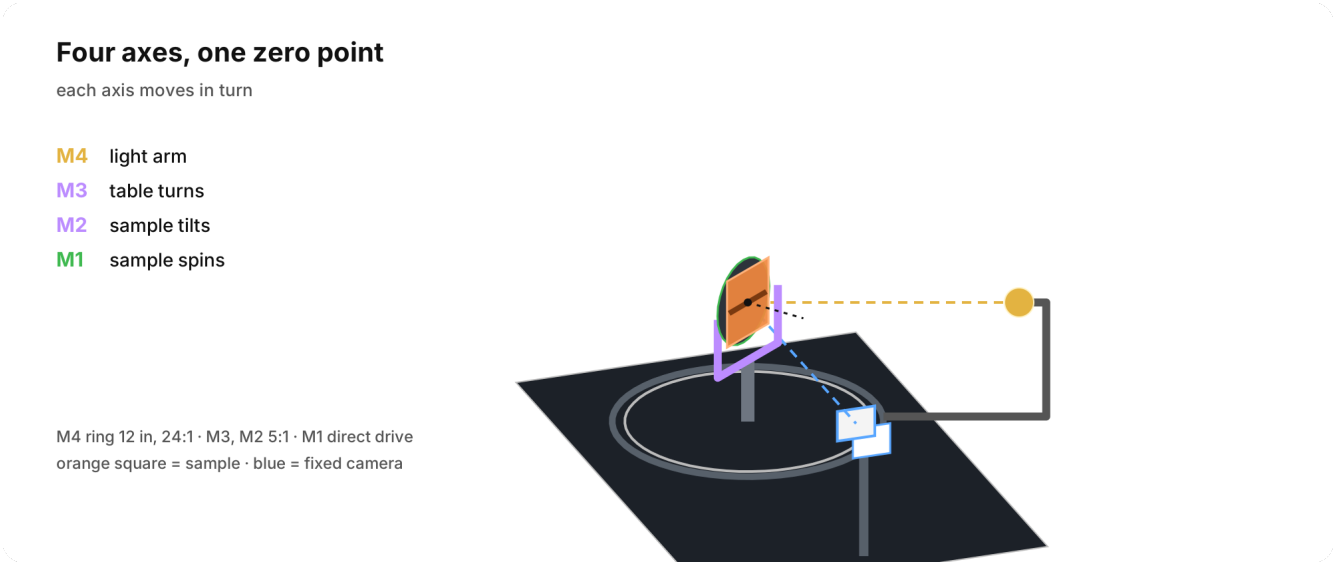


Figure 1: Four axes, one zero point

Axis	Moves	Range	Drive	Step (1/16)	Built in
M4	light arm around the sample	20–324°	GT2 belt on 12 in ring, 24:1	0.0047°	stage 04
M3	table (sample yaw)	±180°	20T → 100T, 5:1	0.0225°	stage 05
M2	sample tilt (yoke)	±90°	20T → 100T, 5:1	0.0225°	stage 08
M1	sample spin (platter)	±180°	direct drive (pancake motor), 1/64 microstep	0.028°	stage 09

Measuring camera fixed on +X at 350 mm; scan camera fixed at 650 mm, 15° up, looking over it, so no camera ever moves or refocuses (one-camera option: move it, $81 cheaper). Light 300 mm from the zero point. Optical homing on every axis, one over-travel switch in the e-stop loop (light arm, at 17°; the other ends rely on soft limits and the e-stop), 30 mm keep-out and 5° minimum light–camera angle (Ansari-Asl et al., 2024).

Why the light is on a side ring. Reflectance depends only on the light and camera directions *relative to the sample*, and the table (M3) and tilt (M2) already turn the sample to every direction an overhead light could reach. A horizontal ring keeps the arm's weight on the bearing (the motor only moves it), stays low and stiff, and keeps cables out of the light path. A light on the camera axis (coaxial, via a beam splitter) sees only back-scatter, not the mirror peak, and loses ~75 % of the light in the splitter; it stays a possible later add-on for retroreflective materials.

Two sensors, two jobs

	Camera (IMX477, fixed, 350 mm)	Spectral sensor (AS7341, above the camera)
Measures	images: brightness per pixel, 3 colours	8 visible bands 415–680 nm + clear + NIR (10 readings), one spot
Used for	angles (mirror test), textures (SVBRDF), 3D scan, calibration	colour spectrum vs angle (spectral BRDF)

	Camera (IMX477, fixed, 350 mm)	Spectral sensor (AS7341, above the camera)
Command	`opengonio capture …`	`opengonio capture … --spectral` (or `--no-camera`)
Stage	06	10 (optional, +$17)

References

Ansari-Asl, M., Barbieri, M., Obein, G., & Hardeberg, J. Y. (2024). Optical and electromechanical design and implementation of an advanced multispectral device to capture material appearance. *Journal of Imaging*, *10*(3), 55. https://doi.org/10.3390/jimaging10030055

Hardware

Every electrical connection, pin by pin (power, SKR Pico, ESP32-S3 header, lamp, USB; step by step in the wiring guide): README → Wiring.

Subsystem	Parts	Stage
Power	24 V 3 A wall adapter with a screw-terminal jack, latching e-stop in its own box, pre-wired limit switch, lever connectors	01
Controller	BTT SKR Pico (grblHAL, USB-C) + ESP32-S3 hub, keypads	02
Frame	printed 600 × 600 mm deck (9 tiles), printed 20 × 20 bars, M2 screws	03
Light arm (M4)	12 in lazy Susan, GT2 belt rim, 24:1	04
Table (M3), tilt (M2), spin (M1)	NEMA 17 steppers, 5:1 belts, M8 bolt axles, direct-drive spin	05 · 08 · 09
Camera	12 MP IMX477 USB, 16 mm M12 lens, printed f/8 stop	06
Light	Citizen CLU701 COB, colour rendering index (CRI) ≥ 90, 2700 K, in a solderless TE holder, 350 mA wire-lead driver, pin-fin heatsink	06

- **No tools beyond a kit:** lever connectors, screw terminals, modules instead of loose chips, M2 screws self-tapping into printed pilots.
- **One screw size:** M2 everywhere; M3 only into the motors, M8 bolts as axles.
- **Printed parts:** black ABS (acrylonitrile butadiene styrene) on an enclosed printer; ≈ 2.7 kg, half of it the deck.
- **Circuit board (future):** a carrier board that replaces the modules and jumpers: pcb/.
- **Optical homing** on every axis; one over-travel switch in the e-stop loop (light arm, at 17°; the other ends rely on soft limits and the e-stop).
- **Cost:** see the bill of materials (pictures, buy links, STEP/STL downloads).

Open-hardware practice (Open Source Hardware Association, n.d.; Pearce, 2012): Appendix E.

References

Open Source Hardware Association. (n.d.). *Open source hardware (OSHW) definition 1.0*. Retrieved https://oshwa.org/definition/

Pearce, J. M. (2012). Building research equipment with free, open-source hardware. *Science*, *337*(6100), 1303–1304. https://doi.org/10.1126/science.1228183

Tools and joins

Nothing is assumed owned except a laptop: the tools are rows `TLS-01`–`TLS-10` with prices, a separate total from the machine's (printer $279 of it; a print service or makerspace replaces it). Prints are black ABS, which needs an enclosed printer and a ventilated room. The parts were picked so nothing needs a soldering iron, drill, saw, crimper or heat-set inserts. Mark them `have` / `ordered` in PARTS.md.

Join	How, without solder or drilling	Parts
Wire to wire	strip 11 mm, lever connector (thin jumper wire: fold the stripped end double)	PWR-20 P47
Wire to terminal	stripped, twisted wire under the screw (2 wires may share one)	PWR-30 jack, PWR-28 e-stop, SKR Pico POWER
Wire to a pin	jumper wire (cut one in half for pin → bare wire); XH leads crimped on both ends for endstops; PH→XH cables for motors	CTL-07 C10, CTL-06, CTL-11 C11
Small electronics	modules with screw terminals (opto PWR-10, MOSFET MSK-03) or wire leads (LED driver LSR-02, into screw terminals and lever connectors), held in printed frames with double-sided tape; no breadboard, no loose chips	PWR-10, MSK-03, LSR-02
Boards	SKR Pico (connectors fitted), ESP32-S3 dev board and Pi Zero 2 WH with headers soldered	CTL-01 C12, CTL-02 C13, STA-01
COB LED	sits in a solderless holder with poke-in wire terminals; the lamp cap clamps it	LSR-06 L23
Printed part to the deck	M2 × 12 into the pilot boss already printed at that spot (deck map); spare pilots every 40 mm	FRM-06
Deck tile to tile	M2 × 10 + nut through the shared walls, 3 per edge	FRM-06
Screw into a printed part	M2 self-taps into 1.7 mm pilots (`M2_TAP`, set from the coupon); one screw size everywhere. Only the motor faces take M3 (FRM-07)	FRM-06
Bar to clamp	printed 20 × 20 bars with M2 pilots at every clamp station; couplers join the halves	FRM-02, FRM-09i
Axles	M8 bolts instead of shafts: the table drum turns on two 608s around a fixed bolt (one nut clamps the inner races); the tilt cradle turns on two stub bolts; spin is direct drive. No shaft to cut, no flat to file	TBL-05, TLT-05, SPN-01

Trade-off: the solderless power path has no reverse-polarity MOSFET or surge diode. The jack's screws are marked + / −, and step M04.2 checks the adapter's polarity at the jack with the meter (M06.1 again at MOT+ / MOT−) before the motor board is connected; the SKR Pico has its own input fuse and capacitors. Short-circuit protection comes from the adapter itself (listing), not from anything built here.

3D files and making parts

Every part has a 3D model in the repo — list and status per part: cad/README.md.

Folder	What
`phases/NN/3d/`	printed parts: **STEP** (Onshape: Insert → Import, editable solid) and **STL** (slicer)
`cad/bought/`	dummy models of every bought part at true size (motors, boards, bearings, belts, camera, lens…)
`cad/open/`	real open-licence models (Adafruit MIT, KiCad CC-BY-SA + exception)
`cad/assembly/opengonio_assembly.step`	the whole machine, one STEP: import this into Onshape to start
`cad/vendor/` (not committed)	real vendor models: `bash scripts/get_cad.sh`
`cad/PRINT.md`	print list: how many of each, size vs build volume, ≈ grams of ABS (core ≈ 2.7 kg, deck tiles ≈ 1.4 kg)

Putting it together: the build book shows every modelled part in order, one step per page (PDF); the order is `docs/assembly.yaml`, the words come from the stage checklists.

`make cad` regenerates dummies + assembly and runs `make cad-check` (no collisions across every axis' travel, clear sight lines from both cameras and the lamp to the sample); `make cad-zip` packs every STEP into `dist/opengonio-cad.zip` (one Onshape import).

- **Starter geometry, generated:** `python tools/cad/parts.py` (CadQuery; parameters at the top). Print the tolerance coupon first and set `CLEAR` / `M2_TAP`.
- **Vendor models:** `bash scripts/get_cad.sh` → `cad/vendor/` (not committed); browser-only downloads: `bash scripts/get_cad.sh --from /mnt/c/Users/aleks/Downloads`.
- **Deck:** deck map (tiles and every pilot; `PLACE` in `tools/cad/parts.py` is the source) · frame design.
- **Wiring:** power netlist · controller pins · standalone wiring.
- **Onshape:** set-up steps — import the STEP files, edit, export STEP + STL back.



Figure 2: All printed parts

Calibration

Quantity	Method	Pass
Angle offset	mirror peak must sit at M3 = M4 / 2 (Ansari-Asl et al., 2024)	residual ≤ 0.1°
Dark level C_d	lamp-off frame	—
Camera response γ	exposure series (Debevec & Malik, 1997)	linear fit
Absolute BRDF	PTFE (polytetrafluoroethylene) tile, ≈ 99 % diffuse, at 45° (Weidner & Hsia, 1981)	—
Lamp drift	repeat the first pose at the end	≤ 2 %

$$f_r = \frac{C_{\text{lin}}}{C_{w,\text{lin}}} \cdot \frac{\rho_w}{\pi} \cdot \frac{\cos 45^\circ}{\cos \theta_i}, \qquad C_{\text{lin}} = (C - C_d)^\gamma$$

C camera counts, C_w white tile, ρ_w its reflectance. Steps: stage 07.

References

Ansari-Asl, M., Barbieri, M., Obein, G., & Hardeberg, J. Y. (2024). Optical and electromechanical design and implementation of an advanced multispectral device to capture material appearance. *Journal of Imaging*, *10*(3), 55. https://doi.org/10.3390/jimaging10030055

Debevec, P., & Malik, J. (1997). Recovering high dynamic range radiance maps from photographs. *Proc. SIGGRAPH '97*. https://pauldebevec.com/Research/HDR/debevec-siggraph97.pdf

Weidner, V. R., & Hsia, J. J. (1981). Reflection properties of pressed polytetrafluoroethylene powder. *Journal of the Optical Society of America*, *71*(7), 856–861. https://doi.org/10.1364/JOSA.71.000856

3D scanning

The camera never moves; the sample axes turn the object, so every photo's pose is known.

1. **Pose from the axes:** camera calibrated once against the turntable (hand-eye, AX = YB) (Shah, 2013).
2. **Quick shape:** silhouettes → visual hull (Laurentini, 1994).
3. **Full mesh:** photogrammetry (Griwodz et al., 2021), then screened Poisson reconstruction (Kazhdan & Hoppe, 2013).
4. **Surface detail (optional LED bank):** photometric stereo from known light directions (Ackermann & Goesele, 2015).
5. **Material:** measured BRDF fitted onto the mesh; export to a game engine (Rendering).

Stages 11 · 12 · 13. Background: Appendix C.

References

Ackermann, J., & Goesele, M. (2015). A survey of photometric stereo techniques. *Foundations and Trends in Computer Graphics and Vision*, *9*. https://doi.org/10.1561/0600000065

Griwodz, C. et al. (2021). AliceVision meshroom: An open-source 3D reconstruction pipeline. *Proc. ACM MMSys*. https://doi.org/10.1145/3458305.3478443

Kazhdan, M., & Hoppe, H. (2013). Screened Poisson surface reconstruction. *ACM Transactions on Graphics*, *32*(3), 29. https://doi.org/10.1145/2487228.2487237

Laurentini, A. (1994). The visual hull concept for silhouette-based image understanding. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *16*(2). https://doi.org/10.1109/34.273735

Shah, M. (2013). Solving the robot-world/hand-eye calibration problem using the Kronecker product. *Journal of Mechanisms and Robotics*, *5*(3), 031007. https://doi.org/10.1115/1.4024473

Rendering

What the machine measures, on a real shape: the fitted material and the scanned object as computer graphics.

Step	Command	File
1. Measure the Bidirectional Reflectance Distribution Function (BRDF)	`opengonio capture inplane` (after `calibrate white`)	`curve.csv`
2. Fit base colour, roughness, metallic	`opengonio fit data/<run>`	`fit.json`, `fit.svg`
3. Scan the object	`opengonio scan run`, `opengonio scan hull`	`hull_mesh.ply`, `hull.glb`
4. Combine shape + material	`opengonio render data/<run> --mesh scan_out/hull_mesh.ply`	`hull_mesh.glb`, `hull_mesh.png`
5. Render	Render page, Blender, Unity, Unreal, three.js	pixels

Without hardware: `opengonio render demo` runs all five on simulated data (`data/render_demo/`).

One model everywhere

The fit is Lambert + GGX with metallic-roughness parameters (Walter et al., 2007) (Khronos Group, n.d.), the model glTF, Blender, Unreal (Karis, 2013) and Unity use. Three renderers draw it with the same formula:

Renderer	Where	Use
`opengonio.render.raster`	numpy, CPU	pictures in scripts and tests; no GPU
`docs/og/render-gl.js`	WebGL 2, browser	Render page and the control app's material preview
Blender, Unity, Unreal, three.js	the `.glb`	production

`tests/test_render.py` checks that the renderer's BRDF equals `fit.py`'s, and that a lit white sphere has the brightness the formula predicts.

The `.glb`

- glTF 2.0 binary (Khronos Group, n.d.): one mesh, one metallic-roughness material, provenance (run, holdout, notes) in `extras`.
- Geometry stays in the scanner's millimetres, Z up; the node's scale (0.001) and rotation make it metres, Y up.
- Unity: smoothness = 1 − roughness (URP Lit).

Lighting

- Lamp: the fitted model, exactly.
- Studio (sky gradient + two softboxes): approximate. A fitted curve lights the shadowed sides (Karis, 2014); it is not measured.
- Tone: ACES filmic (Narkowicz, 2016), then sRGB.

Scan → mesh

The visual hull's boundary voxels become triangles, then λ/μ smoothing removes the voxel steps without shrinking (Taubin, 1995). It is still the visual hull: cavities come out filled (the demo mug: hull 43 cm³ against 17 cm³ real).

Limits

- Grey: one camera channel. A colour tint is typed by hand (`--color R G B`).
- One material per object; no texture.
- Isotropic only.
- Between and beyond the measured angles the picture is the model's guess: check the holdout in `fit.json`.

In the control app

Under a run's curve: **Material**. With a fit it shows it; with absolute values it offers **Fit material**; a relative-only run says why it can't be fitted, and a simulated run previews the simulator's true material. **Download .glb** saves it on a shader ball.

References

Karis, B. (2013). Real shading in Unreal engine 4. *Physically Based Shading in Theory and Practice*. https://cdn2.unrealengine.com/Resources/files/2013SiggraphPresentationsNotes-26915738.pdf

Karis, B. (2014). *Physically based shading on mobile*.

Khronos Group. (n.d.). *glTF 2.0 specification*. Retrieved https://registry.khronos.org/glTF/specs/2.0/glTF-2.0.html#metallic-roughness-material

Narkowicz, K. (2016). *ACES filmic tone mapping curve*.

Taubin, G. (1995). A signal processing approach to fair surface design. *SIGGRAPH*.

Walter, B., Marschner, S., Li, H., & Torrance, K. (2007). Microfacet models for refraction through rough surfaces. *Eurographics Symposium on Rendering*.

Software

Python package `opengonio` (software/). Motion: grblHAL on the BTT SKR Pico over USB-C (FluidNC ESP32 boards also supported); the ESP32-S3 hub reads keypads and sensors.

```
make setup && source .venv/bin/activate
make sim                                    # simulated machine, ends "ALL PASS"
opengonio check-pose M4=60 M3=30            # angles + collision check
opengonio --sim plastic capture inplane --step 10 --out data/sim/plastic
opengonio curve data/sim/plastic            # reflectance curve
```

- Laptop mode: the laptop runs jobs. Standalone mode: a Pi Zero 2 W runs them; `opengonio pull` + `opengonio verify` bring data back.
- `make all`: rebuild docs, checks, tests.

Commands

Try without hardware

```
make setup && source .venv/bin/activate     # once; `source …` again in every new terminal
make all                                     # build + check + tests
make ci                                      # replay the GitHub checks on a clean clone (before p
make sim                                     # simulated machine: mirror → calibrate → dark → gamm
opengonio check-pose M4=60 M3=30             # angles + collision check for one pose
opengonio --sim plastic capture inplane --step 10 --out data/sim/plastic && opengonio curve data/s
opengonio scan demo --out demo               # 3D scan pipeline on a synthetic object
```

`--sim MATERIAL` (mirror, ptfe, matte, paper, plastic) runs any `capture` on a simulated machine; `--sim-error M3=0.8` hides an angle error in it, `--sim-drift 0.3` dims its lamp 0.3 % per shot and `--sim-gamma 2.2` makes its camera non-linear, so you can watch `verify` and `calibrate` find them. The Calibration lab shows the same thing interactively.

Measure and calibrate

Measuring loop: capture, verify, calibrate, curve, fit

```
opengonio capture warmup --minutes 10        # stage 06: lamp settle time -> lights.white.warmup_s
opengonio capture mirror --theta 40          # mirror sweep -> data/mirror_40/ (images + manifest.
opengonio verify data/mirror_40              # PASS / WARN / FAIL: missing images, angles, lamp dr
opengonio curve data/mirror_40               # curve.csv + curve.svg (picture of the run)
opengonio calibrate mirror data/mirror_*     # angle offset -> software/config/calibration.yaml
opengonio capture dark      && opengonio calibrate dark data/dark            # lamp-off level
opengonio capture linearity && opengonio calibrate linearity data/linearity  # camera response (ga
opengonio capture white     && opengonio calibrate white data/white          # PTFE at 45:0 -> abs
opengonio capture inplane --step 10 --out data/paper && opengonio curve data/paper   # brdf_per_sr
opengonio fit data/paper                     # base colour, roughness, metallic -> fit.json (+ Ble
opengonio render data/paper                  # that material on a shader ball -> material.glb + ma
opengonio render data/paper --mesh scan_out/hull_mesh.ply   # on your scanned object -> hull_mesh.
opengonio render demo                        # the whole path on simulated data -> data/render_dem
opengonio capture isotropic --step 15        # stage 08: out-of-plane set from BRDF angles (invers
opengonio capture spin --theta 30            # stage 09: sample spun 0–180° (brushed metal)
opengonio scan hull --out scan_cube --voxel 0.5   # stage 12: silhouettes -> hull.ply + object siz
opengonio scan ps --out scan_ball --sphere        # stage 12: photometric stereo -> normals, error
opengonio calibrate show                     # every value, its date, and what is stale or not mea
```

Mirror run before calibration (`curve.svg`)	Material fit (`fit.svg`)
Mirror sweep: peak 0.4 degrees left of M3 = M4/2	Fitted GGX curves through measured points

Both from the simulated machine (`make build` regenerates them): the mirror peak sits 0.4° left of the orange line until `calibrate mirror` runs; the fit recovers the simulated plastic (base 0.2, roughness 0.3).

Every run shoots its first pose again at the end (`ref_start.png`, `ref_end.png`); `verify` fails the run if the lamp changed by more than `drift_limit_pct` (2 %). Every `manifest.json` carries `schema`, units and the calibration used, so old data can be re-processed later.

Flash and move the motors

Full steps (incl. WSL USB): firmware.md.

```
bash scripts/controller.sh flash          # how to put grblHAL on the SKR Pico (Web Builder .uf2,
bash scripts/controller.sh settings       # send firmware/grblhal/settings.txt over USB-C
bash scripts/controller.sh pins           # output port numbers for machine.yaml lights
bash scripts/e4.sh test                   # home, move, motors off
bash scripts/hub.sh flash oct              # MicroPython onto the ESP32-S3 (oct = N8R8/N16R8 modul
bash scripts/hub.sh                       # button hub firmware
```

Control app: laptop, desktop, phone, website

```
opengonio app --sim mirror                    # browser, simulated machine
opengonio app --window                        # desktop window (pip install "./software[desktop]
opengonio app --lan                           # + phones and tablets on this Wi-Fi: scan the QR
bash scripts/pages.sh                         # once: the website on GitHub Pages (asks before m
python software/packaging/build.py            # desktop app for this computer -> dist/OpenGonio-
```

Each run in the app's list downloads as a zip (↓). The web version runs the same Python in the browser (Pyodide) on the simulated machine. The optional status screen on the hub shows the machine's state without any app: `SCREEN ON` on the hub's serial port (hub add-ons).

Standalone mode (no laptop)

Wiring and setup.

```
bash scripts/station.sh aleks@opengonio.local     # once: installs the station on the Pi Zero 2 W
# on the machine: NEXT = choose job, START = run, STOP = abort
opengonio pull aleks@opengonio.local              # later, on the laptop
opengonio verify data/*                           # PASS / WARN / FAIL per job
```

Publish and update the private GitHub repo (Ubuntu)

```
# 1. Tools
sudo apt update && sudo apt install -y git python3-venv unzip curl
sudo mkdir -p -m 755 /etc/apt/keyrings
curl -fsSL https://cli.github.com/packages/githubcli-archive-keyring.gpg | sudo tee /etc/apt/keyri
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/githubcli-archive-keyring
sudo apt update && sudo apt install -y gh

# 2. Log in (workflow scope lets the CI file push)
gh auth login --hostname github.com --git-protocol https --web --scopes workflow
gh auth setup-git

# 3. Unzip (WSL: /mnt/c/Users/<you>/Downloads)
mkdir -p ~/projects && cd ~/projects
unzip -q ~/Downloads/Gonioreflectometer-Open-Source.zip
cd Gonioreflectometer-Open-Source

# 4. Private repo, push, milestones, issues
bash scripts/publish.sh

# 5. Check
gh repo view --web
```

Other account over SSH: `OWNER=<user> REMOTE=git@<host-alias>:<user>/Gonioreflectometer-Open-Source.git bash scripts/publish.sh`

Every update after that (pushes, then syncs milestones and issues with `.github/phases.yml`):

```
cd ~/projects/Gonioreflectometer-Open-Source
```

```
# your own edits (same as: make publish m="…")
bash scripts/update.sh "M05.1 done"

# a new zip (keeps your history; replaces repo files with the zip's)
bash scripts/update.sh "update from new zip" /mnt/c/Users/<you>/Downloads/Gonioreflectometer-Open-
bash scripts/get_papers.sh
```

Build guide

Start here

1. **Set up once:** publish the repo (commands), then `make setup`. Decide your parts in PARTS.md.
2. **Try the software without hardware:** Try without hardware.
3. **Do the steps in order.** `make next` prints the next one; its stage checklist has the full instructions.
4. **After each step:** `make done id=M05.1` closes its GitHub issue, ticks it in the checklists and figures, and publishes.

Every step (= one GitHub issue) is **Time · Needs → Do** (numbered actions) → **Verify** (a measurement with a pass number) → **If a check fails** (what to look at first) → **Record** (the file to fill in). To change or extend anything, see Tweak and extend.

Two ways to run

	Laptop mode	Standalone mode (stage 15)
Who runs jobs	laptop (`opengonio`)	Pi Zero 2 W (`opengonio station`)
Inputs	laptop or buttons	buttons + OLED menu; phone WebUI for jogging
Data	processed live	saved on the Pi → laptop `opengonio pull` + `opengonio verify`
Extra cost	$0	$38 kit (optional)

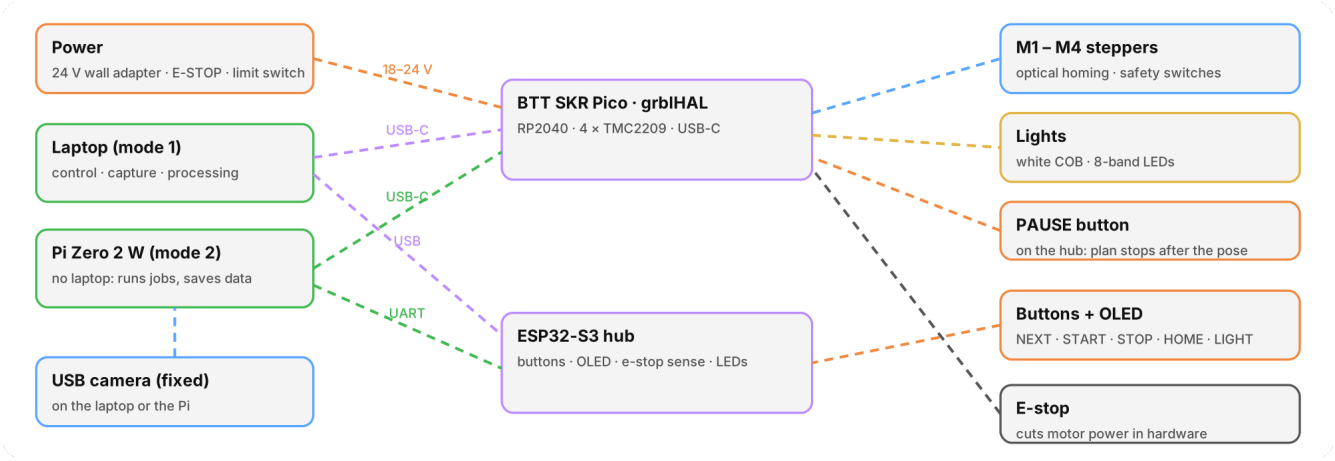


Figure 3: Laptop mode and standalone mode

Theory and equations

- **Bidirectional reflectance distribution function (BRDF):** how much light arriving from one direction leaves toward another. The machine sets both directions: the light arm (M4) sets where light comes from; the fixed camera is where it goes; the sample axes (M3, M2, M1) turn the sample between them.
- **Isotropic vs anisotropic:** most materials look the same when spun about their normal (3 angles). Brushed metal, satin and hair do not (4 angles), so they need M1.
- **Microfacet model:** a rough surface is many tiny mirrors; GGX describes how they are tilted, Fresnel how much each reflects, Smith how they shadow each other. Fitting roughness, colour and metalness to measurements gives a game-engine material.
- **Mirror check:** a mirror only reflects when the incident and reflected angles are equal, so with no tilt the peak must sit at M3 = M4 / 2. That calibrates the angles.
- **Calibration:** the mirror peak gives the angle offset; a lamp-off image gives the dark level; a PTFE tile (≈ 99 % white, diffuse) at 45° turns camera counts into absolute BRDF units; repeating the first pose at the end of every run shows whether the lamp drifted.
- **3D scan:** the camera never moves, so every photo's pose is known from the axes; silhouettes give a quick shape (visual hull), photogrammetry gives the full mesh, and (with the optional LED bank) lights from several known directions give surface detail (photometric stereo).

BRDF and reflection equation (Hanrahan, 2016; Nicodemus et al., 1977)

$$f_r(\omega_i \to \omega_r) = \frac{dL_r(\omega_i \to \omega_r)}{dE_i}\ [\mathrm{sr}^{-1}], \qquad L_r(x,\omega_r) = \int_{H^2} f_r\, L_i(x,\omega_i)\cos\theta_i\, d\omega_i$$

Reciprocity, isotropy, energy (Hanrahan, 2016)

$$f_r(\omega_r \to \omega_i) = f_r(\omega_i \to \omega_r), \qquad f_r^{\mathrm{iso}} = f_r(\theta_i, \theta_r, |\varphi_r - \varphi_i|), \qquad \int_{H^2} f_r \cos\theta_r\, d\omega_r \le 1$$

Anisotropic (needs M1) (Ward, 1992)

$$f_r^{\mathrm{aniso}} = f_r(\theta_i, \varphi_i, \theta_r, \varphi_r) \quad \text{(4 angles; } \varphi \text{ measured from the brushing direction)}$$

Mirror, Snell, total internal reflection (Hanrahan, 2016)

$$\hat{R} = \hat{I} - 2(\hat{I}\cdot\hat{N})\hat{N}, \qquad \eta_i \sin\theta_i = \eta_t \sin\theta_t, \qquad 1 - \mu^2\left(1 - (\hat{I}\cdot\hat{N})^2\right) < 0$$

Fresnel (Schlick), Lambert (Hanrahan, 2016; Schlick, 1994)

$$F(\theta) = F_0 + (1 - F_0)(1 - \cos\theta)^5, \quad F_0 = \left(\tfrac{\eta-1}{\eta+1}\right)^2, \qquad f_{r,d} = \frac{\rho_d}{\pi}$$

Fitted model (Lambert + GGX) (Heitz, 2014; Trowbridge & Reitz, 1975; Walter et al., 2007)

$$f_r = (1-m)(1-F)\frac{\rho}{\pi} + \frac{D\,G_2\,F}{4\cos\theta_i\cos\theta_r}, \quad D = \frac{\alpha^2}{\pi\left((h\cdot n)^2(\alpha^2-1)+1\right)^2}, \quad h = \frac{\omega_i + \omega_r}{\|\omega_i + \omega_r\|}$$

Sample orientation and angles (Ansari-Asl et al., 2024)

$$R = R_z(M_3)\,R_y(-M_2)\,R_x(M_1), \quad \omega^s = R^\top \omega^w, \quad \theta = \arccos(\omega^s \cdot \hat{n}), \quad \varphi = \mathrm{atan2}(\omega_z^s, \omega_y^s)$$

Light and camera directions (Ansari-Asl et al., 2024)

$$\hat{l} = R_z(M_4)\,\hat{x}, \qquad \hat{c} = \hat{x}\ (\text{fixed}), \qquad \omega_i(p) = \frac{P_L - p}{\|P_L - p\|}, \quad \omega_r(p) = \frac{P_C - p}{\|P_C - p\|}$$

In-plane geometry (M2 = M1 = 0) (Ansari-Asl et al., 2024)

$$\theta_r = M_3, \qquad \theta_i = M_4 - M_3, \qquad \text{specular: } M_3 = \tfrac{M_4}{2}$$

Step size

$$\Delta\theta = \frac{360^\circ}{200 \cdot 16 \cdot G} \;\Rightarrow\; 0.0047^\circ\,(M_4,\, G = 24),\; 0.0225^\circ\,(M_3, M_2);\; M_1 \text{ direct at } 1/64\text{: } 0.028^\circ$$

Optics (Ansari-Asl et al., 2024)

$$\frac{1}{p'} - \frac{1}{p} = \frac{1}{f}, \quad m = \frac{p'}{p}, \quad d_{\text{Airy}} = 2.44\,\lambda N, \quad \text{DOF} \approx \frac{2Nc(1+m)}{m^2}$$

HDR (Debevec & Malik, 1997)

$$g(Z_{ij}) = \ln E_i + \ln \Delta t_j$$

Repeatability from the mirror (Ansari-Asl et al., 2024)

$$\Delta\theta_L = \frac{s\, p_{px}\, P}{f\, D_L}$$

Calibration: angle offset, camera response, absolute BRDF, lamp drift (Ansari-Asl et al., 2024; Debevec & Malik, 1997; Hanrahan, 2016)

$$\delta_{M3} \leftarrow \delta_{M3} - \overline{\left(M_3^{\text{peak}} - \tfrac{M_4}{2}\right)}, \qquad \ln(C - C_d) = \tfrac{1}{\gamma}\ln t + k, \qquad C_{\text{lin}} = (C - C_d)^\gamma$$

$$f_r = \frac{C_{\text{lin}}}{C_{w,\text{lin}}} \cdot \frac{\rho_w}{\pi} \cdot \frac{\cos 45^\circ}{\cos\theta_i}, \qquad \text{drift} = \frac{C_{\text{end}} - C_{\text{start}}}{C_{\text{start}} - C_d} \le 2\,\%$$

C = camera counts (centre of the sample), C_d = lamp off, t = exposure, γ = camera response (1 = linear), C_w = PTFE at $\theta_i = 45^\circ$, ρ_w = PTFE reflectance, δ = angle offset (true = commanded + δ). Same exposure for all shots.

Out-of-plane poses from BRDF angles (inverse kinematics, stages 08–09) (Ansari-Asl et al., 2024; Hanrahan, 2016)

$$\cos\alpha = \cos\theta_i\cos\theta_r + \sin\theta_i\sin\theta_r\cos\Delta\varphi, \quad M_4 = \alpha, \quad \hat{n} = a\hat{c} + b\hat{l} + g(\hat{c}\times\hat{l}), \quad M_2 = \arcsin n_z, \quad M_3 = \text{atan2}(n$$

a, b from $\hat{n}\cdot\hat{c} = \cos\theta_r$, $\hat{n}\cdot\hat{l} = \cos\theta_i$; g from $|\hat{n}| = 1$. Light on the camera axis ($\alpha < 20^\circ$) is out of reach, so pure back-scatter is not measured.

3D scan: projection, view elevation (*COLMAP Output Format*, n.d.; *OpenCV 'solvePnP'*, n.d.)

$$\lambda[u,v,1]^\top = K(R_{oc}p_O + t_{oc}), \quad R_{oc} = R_{wc}RP, \qquad \epsilon = 90^\circ - M_2 + e_{cam}$$

3D scan: camera calibration (AX = YB) (Shah, 2013)

$$\min_{X,Y} \sum_i \left\| \log\left(A_i^{-1} X B_i Y\right) \right\|^2$$

3D scan: visual hull, photometric stereo (Laurentini, 1994; Woodham, 1980)

$$V = \bigcap_j \{p : \pi_j(p) \in S_j\}, \qquad I = L\,g,\ g = L^+ I,\ \rho = \|g\|,\ \hat{n} = g/\rho$$

Full notes: theory.md · scan methods.

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Test materials

θi = M4 − M3 (light from the normal), θr = M3 (camera from the normal), in-plane. Isotropic needs 3 angles (M4, M3, M2); anisotropic needs 4 (adds M1).

Material	Type	Light (M4) · camera (θr = M3)	What to look for
PTFE white reference	isotropic, diffuse	45° · 0° (CIE 45°:0°)	flat response; calibrates brightness (Weidner & Hsia, 1981)
Matte paper	isotropic, diffuse	60° · sweep M3	≈ cos θr
First-surface mirror	isotropic, specular	60° · 30°	peak at M3 = M4/2; calibrates angles
Glossy plastic (acrylic, LEGO)	isotropic, glossy	40/80/120° · M4/2 ± 10°	lobe width → roughness
Metallic car paint	isotropic, flake	90° · 35° (15° off specular)	sparkle, colour shift (ASTM International, 2025)
Brushed aluminium	**anisotropic**	60° · 30°, spin M1 0–180°	highlight stretches across the brushing (Ward, 1992)
Satin ribbon	**anisotropic**	70° · 35°, M2 15°, M1 15° steps	two lobes vs direction
Velvet / retroreflective tape	back-scatter	20° (closest allowed) · 10°	peak toward the light
3D scan: matte print, painted toy	object	lamp 30°, scan camera 650 mm / 15° up, M2 85→25°, M1 36 steps	spray shiny parts matte

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Tweak and extend

Everything is generated from a few source files. Edit the source, run `make all`, then `make publish m="…"`.

To change…	Edit	Then
a step, a milestone, a pass number	`.github/phases.yml` (`do`, `verify`, `fail`, `record`, `time`, `wait`, `needs`)	`make all`
mark a step done	—	`make done id=M05.1` (closes the issue, refreshes ticks and figures)
add a stage / milestone / step	`python tools/new.py stage 18-name "Title"` · `milestone 04-light-arm "Title" "Goal"` · `step M14 "Title"`	fill the TODO lines, `make all`
agree / have / ordered / swap / skip a part, what you paid	PARTS.md (also on GitHub in the browser)	`make` locally, or nothing: *parts-sync* runs on GitHub
a part's spec, price or link; a new part	`phases/NN/bom.csv` (or `python tools/new.py part …`)	`make all` (budget is re-checked)
a printed part's size	the PARAMETERS block or the part function in `tools/cad/parts.py`	`make parts`
a printed part, by hand	import `phases/NN/3d/*.step` into Onshape, export STEP + STL back	`make all`
machine geometry, ports, gear ratios (what you **built**)	`software/config/machine.yaml`	`make all` (simulator numbers update too)
angle offsets, dark level, white reference (what you **measured**)	written by `opengonio calibrate …` into `software/config/calibration.yaml`	commit it
lamp warm-up, drift limit	`lights.white.warmup_s` (machine.yaml) · `drift_limit_pct` (calibration.yaml)	—
what you paid	`Paid USD` column in `phases/NN/bom.csv`	`make` prints *spent so far*
mark a stage done	—	`make release s=06` (tag `v0.6` + every STEP/STL as a zip on GitHub)
motor board settings	`tools/grblhal_settings.py` → `firmware/grblhal/settings.txt`	`make build`, then `bash scripts/controller.sh settings`
buttons, screen, spectral sensor	`firmware/hub/main.py`	`bash scripts/hub.sh`
a reference	`python tools/new.py ref G "Source" "Used for (phase NN)"`; cite as `[G53](https://github.com/Normansrule/Gonioreflectometer-Open-Source/blob/main/REFERENCES.md)`	`make all`
how-to text for a stage	`phases/NN/design.md` (how) · `research.md` (why)	`make check`
a test result	`phases/NN/test.csv`	commit

Rules that keep it simple: - **Generated, don't edit by hand:** `phases/NN/README.md`, `phases/NN/3d/README.md`, the tables between `<!-- … -->` markers in this README, `docs/assets/*.svg`, `docs/assets/steps/*.svg`, `docs/machine.js`, `docs/plan.js`, `.github/progress.json`, `optics.md`, `power_budget.md`, `deck-map.svg`. - **Specs vs decisions:** `bom.csv` holds what a part is (spec, price, link); `PARTS.md` holds what you decided (status, paid, note). `make` copies decisions into `bom.csv` (`Status`, `Own`, `Paid USD`), so edit those three only in `PARTS.md`. Zip updates never overwrite your `PARTS.md`. - **Math on GitHub:** `make check` rejects TeX macros GitHub refuses (operator-name and new-command macros); write `\mathrm{atan2}`. - **Built vs measured:** `machine.yaml` holds design values; `calibration.yaml` holds measured ones with dates. `verify` warns when a calibration is older than 30 days. - **Simulator = software:** the web simulator reads `docs/machine.js`, generated from `machine.yaml`; `make check` fails if they differ. - **Test without hardware first:** a code change must keep `make sim` at `ALL PASS`. - **Go back:** `git checkout v0.6` returns to the state when stage 06 passed (`git checkout main` to return). - **Every step has a time and a pass number:** `make check` rejects a step without Do / Verify, with a malformed `time`, or with `TODO` left in it. - **Milestone IDs never change** once published: new ones get the next free number, so GitHub issue history stays attached. - **`make check`** lists every problem in one place: milestone format, parts lists, the budget cap, reference IDs, broken links, machine and calibration config. - **Measurement data** goes in `data/<job>/`: `manifest.json`, `curve.csv` and `spectral.csv` are committed; images stay local.

Repository map

```
README.md            this guide
CLAUDE.md          rules for AI agents: update + build + commit the PDF after every change
PARTS.md             your working parts list: decisions, paid, notes (edit on GitHub)
Makefile                 make help · setup · build · parts · cad · cad-check · cad-
zip · check · test · paper · export · ci · all · sim · next · done · release · publish
REFERENCES.md        487 sources (also listed below)
phases/NN-name/      one folder per stage
  README.md          checklist: Do / Verify / Record (generated)
  design.md          how it is built (wiring, dimensions, notes)
  research.md        why (requirements, methods)
  bom.csv            parts to buy, prices, links
  test.csv           test log you fill in
  3d/                printed parts: .step (Onshape) + .stl (slicer)
cad/           README.md (model per part) · PRINT.md (print list) · bought/ dummies · open/ open models
pcb/               carrier circuit board: reference spec for a future version (README.md)
firmware/                     grblHAL settings (SKR Pico) · FluidNC config (ESP32-
board option) · MicroPython hub (ESP32-S3)
software/            opengonio: control, capture, calibration, simulator, scanning
  config/            machine.yaml (built) · calibration.yaml (measured)
tools/          check.py (health) · new.py (add things) · roadmap, BOM, figures, CAD parts, deck map, c
scripts/         controller.sh · hub.sh · e4.sh · station.sh · publish.sh · update.sh · release.sh · ge
docs/           interactive simulator (index.html, machine.js, plan.js) + figures (assets/, assets/st
exports/         Word, Overleaf, Zotero, Excel, wiki PDF, glTF copies (generated; see exports/README.m
paper/          HardwareX manuscript: opengonio.tex · references.bib (generated) · build/opengonio.p
wiki/           explanations, Bibliography (APA), Appendices A–E · src/ = sources · scripts/publish_wi
papers/              PDFs: open/ (committed) · personal/ (local only)
data/           measurement jobs: manifest.json, curve.csv, spectral.csv (images stay local; data/sim
.github/phases.yml   stages + milestones (source of every checklist and issue)
.github/progress.json  closed steps, pulled from GitHub by `make done` / `make publish`
```

Paper

Manuscript for *HardwareX* (Elsevier), the open-source scientific hardware journal: paper/.

- PDF: paper/build/opengonio.pdf, rebuilt by `make paper` on every change.
- Sections follow the HardwareX hardware-article template; citations APA 7 (`biblatex-apa`), keys = reference IDs.
- Edit elsewhere: Word · Overleaf zip · references for Zotero; copy edits back into `paper/opengonio.tex`.
- Results sections wait for measured data (stages 06–09).

Bibliography

322 scholarly sources, APA 7th edition. `[G12]` = ID in REFERENCES.md and BibTeX key in paper/references.bib. Topic lists with what each source is used for: Appendix A · Appendix B · Appendix C · Appendix D · Appendix E.

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Appendix A · Reflectance measurement and gonioreflectometers

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

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- Schwartz, C., Sarlette, R., Weinmann, M., & Klein, R. (2013). DOME II: A parallelized BTF acquisition system. *Eurographics Workshop on Material Appearance Modeling*. https://doi.org/10.2312/MAM.MAM2013.025-031 — *Camera-array dome comparison (phase 09)* [G50]

Datasets and data formats

- Matusik, W., Pfister, H., Brand, M., & McMillan, L. (2023). *BRDF database – bidirectional reflectance distribution function*. https://doi.org/10.5281/zenodo.8101681 — *Licensed MERL copy for comparison (phase 17)* [G51]
- Lanevski, D. et al. (2022). Machine-readable universal data format for bidirectional reflectance distribution function and BiRDview—an open-source web-based application for viewing and comparing bidirectional reflectance data. *Color Research & Application*. https://doi.org/10.1002/col.22790 — *BRDF export format, viewer (phase 14)* [G52]

Appendix B · Reflectance theory and models

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

Rendering theory and textbooks

- Kajiya, J. T. (1986). The rendering equation. *Proc. SIGGRAPH '86, Computer Graphics*, *20*(4), 143–150. https://doi.org/10.1145/15922.15902 — *Reflection/rendering equation background (phases 00, 07)* [T1]
- Pharr, M., Jakob, W., & Humphreys, G. (2023). *Physically based rendering: From theory to implementation* (4th ed.). https://www.pbr-book.org/4ed/Reflection_Models/Further_Reading — *BRDF, microfacet, sampling reference (phases 07, 08)* [T2]
- Veach, E. (1997). *Robust Monte Carlo methods for light transport simulation*. https://graphics.stanford.edu bw.pdf — *Reciprocity, importance sampling theory (phases 08, 17)* [T3]
- Weyrich, T., Lawrence, J., Lensch, H. P. A., Rusinkiewicz, S., & Zickler, T. (2009). Principles of appearance acquisition and representation. *Foundations and Trends in Computer Graphics and Vision*, *4*(2), 75–191. https://doi.org/10.1561/0600000022 — *Measurement-to-representation overview (phases 00, 08, 09)* [T4]
- Montes, R., & Ureña, C. (2012). *An overview of BRDF models*. https://digibug.ugr.es/bitstream/handle/104 2012-001TR.pdf — *Model catalogue for choosing fits (phases 07, 08)* [T5]
- Nimier-David, M., Vicini, D., Zeltner, T., & Jakob, W. (2019). Mitsuba 2: A retargetable forward and inverse renderer. *ACM Transactions on Graphics*, *38*(6), 203. https://doi.org/10.1145/3355089.3356498 — *Differentiable reference renderer for fit checks (phase 17)* [T6]
- Mitsuba 3 documentation. (n.d.). *BSDFs*. https://mitsuba.readthedocs.io/en/stable/src/generated/plugins_ — *Render fitted vs RGL measured BSDF (phase 17)* [T7]

Classic analytic reflectance models

- Phong, B. T. (1975). Illumination for computer generated pictures. *Communications of the ACM*, *18*(6), 311–317. https://doi.org/10.1145/360825.360839 — *Baseline lobe model for comparisons (phase 07)* [T8]
- Blinn, J. F. (1977). Models of light reflection for computer synthesized pictures. *Proc. SIGGRAPH '77*. https://doi.org/10.1145/563858.563893 — *Half-vector lobe, legacy export mapping (phase 07)* [T9]
- Torrance, K. E., & Sparrow, E. M. (1967). Theory for off-specular reflection from roughened surfaces. *Journal of the Optical Society of America*, *57*(9), 1105–1114. https://doi.org/10.1364/JOSA.57.00110 — *Microfacet theory, off-specular peak (phase 08)* [T10]
- Trowbridge, T. S., & Reitz, K. P. (1975). Average irregularity representation of a rough surface for ray reflection. *Journal of the Optical Society of America*, *65*(5), 531–536. https://doi.org/10.1364/JOSA.65.000531 — *Origin of GGX/TR distribution (phase 07)* [T11]
- Smith, B. G. (1967). Geometrical shadowing of a random rough surface. *IEEE Transactions on Antennas and Propagation*, *15*(5), 668–671. https://ui.adsabs.harvard.edu/abs/1967ITAP...15..668S/abstra — *Smith masking-shadowing term (phases 07, 08)* [T12]
- Oren, M., & Nayar, S. K. (1994). Generalization of Lambert's reflectance model. *Proc. SIGGRAPH '94*. https://doi.org/10.1145/192161.192213 — *Rough-diffuse alternative to Lambert (phases 07, 08)* [T13]
- Lafortune, E. P. F., Foo, S.-C., Torrance, K. E., & Greenberg, D. P. (1997). Non-linear approximation of reflectance functions. *Proc. SIGGRAPH '97*. https://doi.org/10.1145/258734.258801 — *Multi-lobe fit, retroreflection (phase 08)* [T14]
- Ashikhmin, M., & Shirley, P. (2000). An anisotropic Phong BRDF model. *Journal of Graphics Tools*, *5*(2), 25–32. https://doi.org/10.1080/10867651.2000.10487522 — *Anisotropic fit alternative (phase 09)* [T15]
- Rusinkiewicz, S. (1998). A new change of variables for efficient BRDF representation. *Rendering Techniques '98*. https://doi.org/10.1007/978-3-7091-6453-2_2 — *Half/difference angle*

data layout (phases 08, 17) [T16]

Microfacet theory and sampling (modern)

- Heitz, E. (2014). Understanding the masking-shadowing function in microfacet-based BRDFs. *Journal of Computer Graphics Techniques*, *3*(2), 48–107. https://jcgt.org/published/0003/02/03/paper.pdf — *Correct Smith G for GGX fit (phases 07, 08)* [T17]
- Heitz, E., & d'Eon, E. (2014). Importance sampling microfacet-based BSDFs using the distribution of visible normals. *Computer Graphics Forum*, *33*(4), 103–112. https://doi.org/10.1111/cgf.12417 — *Sampling fitted GGX in renders (phases 14, 17)* [T18]
- Heitz, E. (2018). Sampling the GGX distribution of visible normals. *Journal of Computer Graphics Techniques*, *7*(4), 1–13. https://jcgt.org/published/0007/04/01/paper.pdf — *Simple VNDF sampler code (phase 14)* [T19]
- Dupuy, J., & Benyoub, A. (2023). Sampling visible GGX normals with spherical caps. *Computer Graphics Forum*, *42*(8). https://doi.org/10.1111/cgf.14867 — *Faster VNDF sampling (phase 14)* [T20]
- Heitz, E., Hanika, J., d'Eon, E., & Dachsbacher, C. (2016). Multiple-scattering microfacet BSDFs with the Smith model. *ACM Transactions on Graphics*, *35*(4), 58. https://doi.org/10.1145/2897824.292 — *Energy loss at high roughness (phase 08)* [T21]
- Turquin, E. (2019). *Practical multiple scattering compensation for microfacet models*. https://blog.selfshadow.com/publications/turquin/ms_comp_final.pdf — *Energy compensation when fitting rough metals (phase 07)* [T22]
- Bagher, M. M., Soler, C., & Holzschuch, N. (2012). Accurate fitting of measured reflectances using a shifted gamma micro-facet distribution. *Computer Graphics Forum*, *31*(4). https://inria.hal.science/hal-00702304 — *Long-tail NDF fitting option (phase 17)* [T23]
- Ribardière, M., Bringier, B., Meneveaux, D., & Simonot, L. (2017). STD: Student's t-distribution of slopes for microfacet based BSDFs. *Computer Graphics Forum*, *36*(2). https://doi.org/10.1111/cgf.13137 — *Generalised GGX tail fitting (phase 17)* [T24]

Fitting measured data and evaluating fits

- Löw, J., Kronander, J., Ynnerman, A., & Unger, J. (2012). BRDF models for accurate and efficient rendering of glossy surfaces. *ACM Transactions on Graphics*, *31*(1), 9. https://doi.org/10.1145/2077341.2077350 — *Fit error metric, glossy model choice (phases 07, 08)* [T25]
- Dupuy, J., Heitz, E., Iehl, J.-C., Poulin, P., & Ostromoukhov, V. (2015). Extracting microfacet-based BRDF parameters from arbitrary materials with power iterations. *Computer Graphics Forum*, *34*(4), 21–30. https://doi.org/10.1111/cgf.12675 — *Robust NDF extraction for GGX fit (phases 08, 09)* [T26]
- Lavoué, G., Bonneel, N., Farrugia, J.-P., & Soler, C. (2021). Perceptual quality of BRDF approximations: Dataset and metrics. *Computer Graphics Forum*, *40*(2). https://doi.org/10.1111/cgf.142636 — *Perceptual fit-quality metric (phases 07, 17)* [T27]
- Kavoosighafi, B. et al. (2025). Perceived quality of BRDF models. *Computer Graphics Forum*. https://doi.org/10.1111/cgf.70162 — *Which model looks best when fitted (phase 17)* [T28]
- Sztrajman, A., Rainer, G., Ritschel, T., & Weyrich, T. (2021). Neural BRDF representation and importance sampling. *Computer Graphics Forum*, *40*. https://doi.org/10.1111/cgf.14335 — *Neural fit stretch; remap to analytic (phase 17)* [T29]

Fresnel and optical constants

- Born, M., & Wolf, E. (2019). *Principles of optics* (7th ed.). https://www.cambridge.org/9781108477437 — *Exact Fresnel equations, polarisation (phases 07, 10)* [T30]
- Gulbrandsen, O. (2014). Artist friendly metallic Fresnel. *Journal of Computer Graphics Techniques*, *3*(4), 64–72. https://jcgt.org/published/0003/04/03/paper.pdf — *Map n,k to reflectivity/edge tint (phase 07)* [T31]

- Polyanskiy, M. N. (2024). Refractiveindex.info database of optical constants. *Scientific Data*, *11*, 94. https://doi.org/10.1038/s41597-023-02898-2 — *Metal F0 sanity check vs mirror, chrome (phases 06, 07)* [T32]

Production shading models (course notes)

- Burley, B. (2012). Physically-based shading at Disney. *Practical Physically Based Shading in Film and Game Production*. https://static1.squarespace.com/static/58586fa5ebbd1a60e7d76d3e/t/593a3a — *Principled parameters, MERL-based design (phases 07, 17)* [T33]
- Burley, B. (2015). *Extending the Disney BRDF to a BSDF with integrated subsurface scattering*. https://blog.selfshadow.com/publications/s2015-shading-course/burley/s2015_pbs_disney_bsdf_notes.pdf — *Anisotropic alpha mapping, sheen (phase 09)* [T34]
- Karis, B. (2013). Real shading in Unreal engine 4. *Physically Based Shading in Theory and Practice*. https://cdn2.unrealengine.com/Resources/files/2013SiggraphPresentationsNotes-26915738.pdf — *UE GGX/Smith/Schlick, α = roughness² (phase 07)* [T35]
- Lagarde, S., & Rousiers, C. de. (2014). *Moving Frostbite to physically based rendering 3.0*. https://seblagarde.wordpress.com/wp-content/uploads/2015/07/course_notes_moving_frostbite_to_pbr_v3 — *Validation against references, photometric units (phase 07)* [T36]
- Kulla, C., & Conty, A. (2017). *Revisiting physically based shading at Imageworks*. https://blog.selfshadow.com/publications/s2017-shading-course/imageworks/s2017_pbs_imageworks_slid — *Energy compensation, Charlie sheen (phases 07, 09)* [T37]
- Hoffman, N. (2015). Physics and math of shading. *Physically Based Shading in Theory and Practice*. https://blog.selfshadow.com/publications/s2015-shading-course/hoffman/s2015_pbs_physics_ma — *Plain-language theory for docs (phases 00, 07)* [T38]
- Guy, R., & Agopian, M. (n.d.). *Physically based rendering in Filament*. https://google.github.io/filament/Fila — *Readable real-time BRDF derivations (phases 07, 14)* [T39]

Material exchange formats and engine docs

- Khronos Group. (n.d.). *glTF 2.0 specification*. https://registry.khronos.org/glTF/specs/2.0/glTF-2.0.html#metallic-roughness-material — *Core export target and BRDF (phases 07, 13)* [T40]
- *Khronos, KHR_materials_anisotropy*. (n.d.). https://github.com/KhronosGroup/glTF/blob/main/extensions — *Export fitted anisotropy (phase 09)* [T41]
- *Khronos, KHR_materials_sheen, KHR_materials_clearcoat*. (n.d.). https://github.com/KhronosGroup/glTF/ — *Fabric and coated-sample export (phase 09)* [T42]
- *Khronos, KHR_materials_specular, KHR_materials_ior*. (n.d.). https://github.com/KhronosGroup/glTF/blob — *Export measured dielectric F0 (phase 07)* [T43]
- *Khronos, KHR_texture_transform*. (n.d.). https://github.com/KhronosGroup/glTF/blob/main/extensions/2.0 — *SVBRDF texture tiling on models (phases 09, 13)* [T44]
- *Khronos, glTF sample viewer*. (n.d.). https://github.com/KhronosGroup/glTF-Sample-Viewer — *Reference check of exported .glb (phases 07, 13)* [T45]
- Epic Games. (n.d.). *Physically based materials in Unreal engine*. https://dev.epicgames.com/documentation/en-us/unreal-engine/physically-based-materials-in-unreal-engine — *Unreal export mapping (phases 07, 13)* [T46]
- *Unity, HDRP "lit shader"*. (n.d.). https://docs.unity3d.com/Packages/com.unity.render-pipelines.high-definition@14.0/manual/Lit-Shader.html — *Unity export mapping (phases 07, 13)* [T47]
- Blender Manual. (n.d.). *Principled BSDF*. https://docs.blender.org/manual/en/latest/render/shader_nodes/ — *Check exports in Blender (phases 07, 13)* [T48]
- Georgiev, I., Portsmouth, J., Andersson, Z., Herubel, A., King, A., Ogaki, S., & Servant, F. (n.d.). *Autodesk standard surface*. https://autodesk.github.io/standard-surface/ — *DCC/film material mapping (phase 07)* [T49]
- Andersson, Z., Edmondson, P., Guertault, J., Herubel, A., Kutz, P., Machizaud, A., Portsmouth, J., Servant, F., & Stone, J. (2025). OpenPBR surface: An open shading model for physically based materials. *DigiPro 2025*. https://doi.org/10.1145/3744199.3744632 — *Future-proof export, F82-tint metals (phases 07, 16)* [T50]

- *ASWF, MaterialX specification and physically based shading nodes*. (n.d.). https://github.com/AcademySof — *Renderer-neutral material file (phases 13, 16)* [T51]

Spectral rendering

- Smits, B. (1999). An RGB-to-spectrum conversion for reflectances. *Journal of Graphics Tools*, *4*(4), 11–22. https://doi.org/10.1080/10867651.1999.10487511 — *Baseline RGB↔spectrum for 8 bands (phase 10)* [T52]
- Jakob, W., & Hanika, J. (2019). A low-dimensional function space for efficient spectral upsampling. *Computer Graphics Forum*, *38*(2), 147–155. https://doi.org/10.1111/cgf.13626 — *Smooth spectra from fitted RGB albedo (phase 10)* [T53]
- Peters, C., Merzbach, S., Hanika, J., & Dachsbacher, C. (2019). Using moments to represent bounded signals for spectral rendering. *ACM Transactions on Graphics*, *38*(4), 136. https://doi.org/10.1145/3306346.3322964 — *Compact storage of 8-band reflectance (phase 10)* [T54]
- Wilkie, A., Nawaz, S., Droske, M., Weidlich, A., & Hanika, J. (2014). Hero wavelength spectral sampling. *Computer Graphics Forum*, *33*(4), 123–131. https://doi.org/10.1111/cgf.12419 — *Spectral preview render of measured bands (phases 10, 14)* [T55]
- Taubin, G. (1995). A signal processing approach to fair surface design. *SIGGRAPH*. — *λ/μ mesh smoothing without shrinkage: scanned visual hull → mesh (`render/mesh.py`)* [T56]
- Karis, B. (2014). *Physically based shading on mobile*. — *Analytic environment-BRDF fit for the viewer's approximate studio light (`render/raster.py`, `render-gl.js`)* [T57]
- Narkowicz, K. (2016). *ACES filmic tone mapping curve*. — *Tone curve shared by the software and WebGL renderers* [T58]

Appendix C · Calibration, HDR, photometric stereo, 3D scanning

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

3D photo scanner

- *COLMAP output format*. (n.d.). https://colmap.github.io/format.html — *COLMAP text model, quaternions* [S1]
- COLMAP FAQ. (n.d.). *Reconstruct sparse/dense model from known camera poses*. https://colmap.github.io — *run_colmap.sh command sequence, meshing* [S2]
- Schönberger, J. L., & Frahm, J.-M. (2016). *Structure-from-motion revisited*. https://doi.org/10.1007/978-3-319-46487-9_31 — *Photogrammetry engine* [S3]
- Woodham, R. J. (1980). Photometric method for determining surface orientation from multiple images. *Optical Engineering*, *19*(1). https://doi.org/10.1117/12.7972479 — *Photometric stereo* [S4]
- Laurentini, A. (1994). The visual hull concept for silhouette-based image understanding. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *16*(2). https://doi.org/10.1109/34.273735 — *Visual hull* [S5]
- Szeliski, R. (1993). Rapid octree construction from image sequences. *CVGIP: Image Understanding*, *58*(1). https://www.sciencedirect.com/science/article/abs/pii/S1049966083710296 — *Voxel carving* [S6]
- Kutulakos, K. N., & Seitz, S. M. (2000). A theory of shape by space carving. *International Journal of Computer Vision*. https://doi.org/10.1023/A:1008191222954 — *Space carving* [S7]
- *MVTec HALCON 'photometric_stereo' reference*. (n.d.). https://www.mvtec.com/doc/halcon/13/en/photom — *≥ 3 images, non-coplanar light directions* [S8]
- Fitzgibbon, A. W., Cross, G., & Zisserman, A. (1998). *Automatic 3D model construction for turn-table sequences*. https://doi.org/10.1007/3-540-49437-5_11 — *Turntable (rotating-object) reconstruction* [S9]
- *Meshroom / AliceVision*. (n.d.). https://alicevision.org/ — *Alternative reconstruction* [S10]
- *OpenCV 'solvePnP'*. (n.d.). https://docs.opencv.org/4.13.0/d5/d1f/calib3d_solvePnP.html — *Camera axis convention, pose checks* [S11]
- *OpenCV ChArUco calibration*. (n.d.). https://docs.opencv.org/4.13.0/da/d13/tutorial_aruco_calibration.htm — *Scan-mode intrinsics, scanning mat* [S12]
- Schnieders, D., & Wong, K.-Y. K. (2013). Camera and light calibration from reflections on a sphere. *Computer Vision and Image Understanding*. https://www.sciencedirect.com/science/article/abs/pi — *Chrome-ball light calibration* [S13]
- Yu, Chen, & Da. (2025). Geometric calibration for photometric stereo under near-field point light sources using spheres. *Optics & Laser Technology*, *180*. https://www.sciencedirect.com/science/artic — *Near-light correction (phase 11)* [S14]
- Goldman, D. B., Curless, B., Hertzmann, A., & Seitz, S. M. (2005). *Shape and spatially-varying BRDFs from photometric stereo*. https://doi.org/10.1109/ICCV.2005.219 — *Material on the mesh (phase 13)* [S15]
- *Blender manual: PLY, OBJ, glTF 2.0, formats*. (n.d.). https://docs.blender.org/manual/en/latest/files/import — *Mesh import check* [S16]
- *OpenMVS*. (n.d.). https://github.com/cdcseacave/openMVS — *Optional dense/mesh stage* [S17]
- *RealityScan (formerly RealityCapture)*. (n.d.). https://www.realityscan.com/license — *Optional reconstruction* [S18]
- *QuakeHold! Museum putty*. (n.d.). https://www.sosproducts.com/quakehold-museum-putty-p/64810.htm — *Heavier-object option* [S19]
- *Loctite fun-tak 2 oz*. (n.d.). https://www.staples.com/Loctite-Fun-Tak-2-oz-Mounting-Adhesive-Putty-1270884/product_1480018 — *$3.49 object mount* [S20]

Geometric camera calibration

- Zhang, Z. (2000). A flexible new technique for camera calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *22*(11), 1330–1334. https://doi.org/10.1109/34.888718 — *Planar-target intrinsics (phase 06)* `[V1]`
- Tsai, R. Y. (1987). A versatile camera calibration technique for high-accuracy 3D machine vision metrology using off-the-shelf TV cameras and lenses. *IEEE Journal of Robotics and Automation*, *3*(4), 323–344. https://doi.org/10.1109/JRA.1987.1087109 — *Classic calibration background (phase 06)* `[V2]`
- Heikkilä, J., & Silvén, O. (1997). A four-step camera calibration procedure with implicit image correction. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 1106–1112. https://doi.org/10.1109/CVPR.1997.609468 — *Distortion model and correction (phase 06)* `[V3]`
- Brown, D. C. (1971). Close-range camera calibration. *Photogrammetric Engineering*, *37*(8), 855–866. https://www.semanticscholar.org/paper/Close-Range-Camera-Calibration-Brown/1150007b62a3c7dac99c2c8f85c63bfab74891af — *Radial/tangential lens distortion (phase 06)* `[V4]`
- Bouguet, J.-Y. (2022). *Camera calibration toolbox for Matlab*. https://doi.org/10.22002/D1.20164 — *Reference calibration workflow (phase 06)* `[V5]`
- Schöps, T., Larsson, V., Pollefeys, M., & Sattler, T. (2020). Why having 10,000 parameters in your camera model is better than twelve. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. https://openaccess.thecvf.com/content_CVPR_2020/html/Schops_Why_Havi — *Calibration accuracy limits (phase 17)* `[V6]`
- Garrido-Jurado, S., Muñoz-Salinas, R., Madrid-Cuevas, F. J., & Marín-Jiménez, M. J. (2014). Automatic generation and detection of highly reliable fiducial markers under occlusion. *Pattern Recognition*, *47*(6), 2280–2292. https://doi.org/10.1016/j.patcog.2014.01.005 — *ArUco markers on ChArUco board (phases 06, 11)* `[V7]`
- Hartley, R., & Zisserman, A. (2004). *Multiple view geometry in computer vision* (2nd ed.). https://doi.org/10.1017/CBO9780511811685 — *Projection, triangulation, bundle adjustment (phases 06, 11)* `[V8]`
- Szeliski, R. (2022). *Computer vision: Algorithms and applications* (2nd ed.). https://doi.org/10.1007/978-3-030-34372-9 — *General CV reference (phases 06, 07, 08, 09, 10, 11, 12, 13)* `[V9]`

Hand-eye and robot-world calibration (AX=XB, AX=YB)

- Tsai, R. Y., & Lenz, R. K. (1989). A new technique for fully autonomous and efficient 3D robotics hand/eye calibration. *IEEE Transactions on Robotics and Automation*, *5*(3), 345–358. https://doi.org/10.1109/70.34770 — *Hand-eye AX=XB baseline (phase 11)* `[V10]`
- Park, F. C., & Martin, B. J. (1994). Robot sensor calibration: Solving AX=XB on the Euclidean group. *IEEE Transactions on Robotics and Automation*, *10*(5), 717–721. https://doi.org/10.1109/70.326576 — *Lie-group hand-eye solver (phase 11)* `[V11]`
- Daniilidis, K. (1999). Hand-eye calibration using dual quaternions. *International Journal of Robotics Research*, *18*(3), 286–298. https://doi.org/10.1177/02783649922066213 — *Simultaneous rotation/translation hand-eye (phase 11)* `[V12]`
- Zhuang, H., Roth, Z. S., & Sudhakar, R. (1994). Simultaneous robot/world and tool/flange calibration by solving homogeneous transformation equations of the form AX=YB. *IEEE Transactions on Robotics and Automation*, *10*(4), 549–554. https://doi.org/10.1109/70.313105 — *AX=YB formulation (phase 11)* `[V13]`
- Dornaika, F., & Horaud, R. (1998). Simultaneous robot-world and hand-eye calibration. *IEEE Transactions on Robotics and Automation*, *14*(4), 617–622. https://doi.org/10.1109/70.704233 — *AX=YB nonlinear solution (phase 11)* `[V14]`
- Shah, M. (2013). Solving the robot-world/hand-eye calibration problem using the Kronecker product. *Journal of Mechanisms and Robotics*, *5*(3), 031007. https://doi.org/10.1115/1.4024473 — *Closed-form AX=YB (OpenCV SHAH) (phase 11)* `[V15]`
- Tabb, A., & Ahmad Yousef, K. M. (2017). Solving the robot-world hand-eye(s) calibration problem with iterative methods. *Machine Vision and Applications*. https://doi.org/10.1007/s00138-017-0841-7 — *Reprojection-error AX=YB refinement (phase 11)* `[V16]`

- Ali, I., Suominen, O., Gotchev, A., & Ruiz Morales, E. (2019). Methods for simultaneous robot-world-hand–eye calibration: A comparative study. *Sensors*, *19*(12), 2837. https://doi.org/10.3390/s19122837 — *Choosing an AX=YB method (phase 11)* [V17]
- *OpenCV, camera calibration and 3D reconstruction ('calibrateHandEye', 'calibrateRobotWorldHandEye')*. (n.d.). https://docs.opencv.org/4.13.0/d9/d0c/group__calib3d.html — *Hand-eye implementation (phase 11)* [V18]

Radiometric calibration and HDR capture

- Mitsunaga, T., & Nayar, S. K. (1999). Radiometric self calibration. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 374–380. https://doi.org/10.1109/CVPR.1999.78 — *Polynomial response curve (phase 07)* [V19]
- Robertson, M. A., Borman, S., & Stevenson, R. L. (2003). Estimation-theoretic approach to dynamic range enhancement using multiple exposures. *Journal of Electronic Imaging*, *12*(2), 219–228. https://doi.org/10.1117/1.1557695 — *Weighted HDR merge (OpenCV Robertson) (phase 07)* [V20]
- Grossberg, M. D., & Nayar, S. K. (2003). What is the space of camera response functions? *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. https://doi.org/10.1109/TPAMI.2004.88 — *Response-curve prior (EMoR) (phase 07)* [V21]
- Hasinoff, S. W., Durand, F., & Freeman, W. T. (2010). Noise-optimal capture for high dynamic range photography. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 553–560. https://doi.org/10.1109/CVPR.2010.5540167 — *Choosing exposure brackets (phase 07)* [V22]
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Photometric stereo

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Multi-view reconstruction, visual hulls, meshing

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- Furukawa, Y., & Ponce, J. (2010). Accurate, dense, and robust multiview stereopsis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *32*(8), 1362–1376. https://doi.org/10.1109/TPAMI.2009.161 — *Patch-based MVS (PMVS) (phase 12)* `[V38]`
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Reflectance capture rigs and polarization

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- Wolff, L. B. (1990). Polarization-based material classification from specular reflection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *12*(11), 1059–1071. https://doi.org/10.1109/34.61705 — *Polarization physics background (phase 10)* [V52]

Open-source turntable scanners

- *OpenScan (Raspberry Pi photogrammetry scanner)*. (n.d.). https://openscan.eu/pages/openscan-mini — *DIY scanner prior art (phase 11)* [V53]
- *FabScan, media computing group, RWTH aachen university*. (n.d.). https://hci.rwth-aachen.de/fabscan — *Turntable laser scanner prior art (phase 11)* [V54]
- *BQ ciclop scanner + horus software*. (n.d.). https://github.com/bqlabs/ciclop — *Turntable calibration UX reference (phases 11, 14)* [V55]

Texturing and UV unwrapping

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- *Xatlas, mesh parameterization / UV unwrapping library*. (n.d.). https://github.com/jpcy/xatlas — *Automatic UV atlas (phase 13)* [V57]
- Waechter, M., Moehrle, N., & Goesele, M. (2014). Let there be color! Large-scale texturing of 3D reconstructions. *European Conference on Computer Vision*. https://doi.org/10.1007/978-3-319-10602-1_54 — *View selection for mesh textures (phase 13)* [V58]

Appendix D · Standards, colour, optics, safety

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

Appearance and scatter measurement standards (ASTM)

- ASTM International. (2019). *Standard practice for goniometric optical scatter measurements*. https://store.astm.org/e2387-19.html — *BRDF/BTDF measurement practice (phases 07, 08)* [N1]
- ASTM International. (2003). *Standard practice for angle resolved optical scatter measurements on specular or diffuse surfaces*. https://www.astm.org/DATABASE.CART/WITHDRAWN/E1392.htm — *Historical BRDF practice cited by Labsphere (phase 07)* [N2]
- ASTM International. (2005). *Standard practice for goniophotometry of objects and materials*. https://store.astm.org/e0167-96.html — *Goniophotometer geometry conventions (phase 06)* [N3]
- ASTM International. (n.d.). *Standard test method for specular gloss*. https://store.astm.org/standards/d52 — *20/60/85° gloss cross-check (phase 07)* [N4]
- ASTM International. (2023b). *Standard test methods for measurement of gloss of high-gloss surfaces by abridged goniophotometry*. https://store.astm.org/standards/e430 — *Haze/distinctness-of-image near specular (phases 08, 17)* [N5]
- ASTM International. (2025a). *Standard terminology of appearance*. https://doi.org/10.1520/E0284-25 — *Docs glossary terms (phase 00)* [N6]
- ASTM International. (2026). *Standard practice for specifying the geometry of multiangle spectrophotometers*. https://store.astm.org/standards/e2175 — *Reporting light/camera angle geometry (phase 08)* [N7]
- ASTM International. (2025b). *Standard test method for multiangle color measurement of metal flake pigmented materials*. https://doi.org/10.1520/E2194-14R25 — *Aspecular angles for metallic samples (phases 08, 09)* [N8]
- ASTM International. (2023a). *Standard practice for obtaining spectrometric data for object-color evaluation*. https://store.astm.org/standards/e1164 — *Spectral data and calibration practice (phase 10)* [N9]
- *ISO, "paints and varnishes*. (n.d.). https://www.iso.org/standard/56807.html — *Gloss reference geometry (phase 07)* [N10]
- Nadal, M. E., & Thompson, E. A. (2001). NIST reference goniophotometer for specular gloss measurements. *Journal of Coatings Technology*, *73*(917), 73–80. https://www.paint.org/wp-content/uploads/2021/09/jctJUNE01-Nadal.pdf — *Reference-instrument design and uncertainty (phases 06, 17)* [N11]

CIE colorimetry, appearance and colour rendering

- CIE. (2006). *A framework for the measurement of visual appearance*. https://cie.co.at/publications/framew measurement-visual-appearance — *Colour/gloss/translucency/texture scope (phase 00)* [N12]
- CIE. (2018). *Colorimetry, 4th edition*. https://doi.org/10.25039/TR.015.2018 — *XYZ, CIELAB, illuminants from spectra (phase 10)* [N13]
- CIE. (2020). *ILV: International lighting vocabulary, 2nd edition*. https://doi.org/10.25039/S017.2020 — *Radiometric/photometric terminology (phase 00)* [N14]
- CIE. (1995). *Method of measuring and specifying colour rendering properties of light sources*. https://cie.co.at/publications/method-measuring-and-specifying-colour-rendering-properties-light-sources — *COB CRI (Ra) meaning (phases 04, 10)* [N15]
- CIE. (2017). *CIE 2017 colour fidelity index for accurate scientific use*. https://doi.org/10.25039/TR.224.201 — *Rf for white and LED-bank spectra (phase 10)* [N16]
- *K. A. G. Smet, CRI_CIE_rf_2017 calculator (MATLAB/excel)*. (n.d.). https://github.com/ksmet1977/CRI_CIE — *Checking our Rf implementation (phase 10)* [N17]
- IES. (n.d.). *Technical memorandum: IES method for evaluating light source color rendition*. https://store.ies.org/product/technical-memorandum-ies-method-for-evaluating-light-source-

color-rendition/?v=eb65bcceaa5f — *Rf/Rg report for light sources (phase 10)* [N18]
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- Ohno, Y. (2005). Spectral design considerations for white LED color rendering. *Optical Engineering*, *44*(11). https://doi.org/10.1117/1.2130694 — *Designing the 8-band LED spectrum (phase 10)* [N20]

Colour science books

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- Fairchild, M. D. (2013). *Color appearance models* (3rd ed.). https://doi.org/10.1002/9781118653128 — *Appearance/white point for renders (phases 10, 13)* [N22]
- Hunt, R. W. G., & Pointer, M. R. (2011). *Measuring colour* (4th ed.). https://doi.org/10.1002/978111997559 — *Practical colour measurement geometry (phase 10)* [N23]

Camera characterization and colour encoding

- EMVA. (2021). *Standard 1288: Standard for characterization of image sensors and cameras*. https://www.emva.org/standards-technology/emva-1288/ — *IMX477 linearity, noise, dark current tests (phases 06, 07)* [N24]
- *ISO, "photography*. (n.d.-a). https://www.iso.org/standard/79169.html — *Slanted-edge SFR of 16 mm lens (phase 06)* [N25]
- *ISO, "photography*. (n.d.-b). https://www.iso.org/standard/82233.html — *Noise and dynamic range for HDR (phase 07)* [N26]
- *ISO, "graphic technology and photography*. (n.d.). https://www.iso.org/standard/35835.html — *Camera spectral sensitivity/colour matrix (phase 10)* [N27]
- *IEC, "multimedia systems and equipment*. (n.d.). https://webstore.ansi.org/standards/iec/iec61966ed1999 — *sRGB transfer function for texture export (phases 07, 13)* [N28]
- *M. Stokes, M. Anderson, S. Chandrasekar, R. Motta, "a standard default color space for the internet*. (n.d.). https://www.w3.org/Graphics/Color/sRGB.html — *Linear vs gamma-encoded texture handling (phase 13)* [N29]

Reflectance standards (PTFE / Spectralon)

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- Weidner, V. R., Hsia, J. J., & Adams, B. (1985). Laboratory intercomparison study of pressed polytetrafluoroethylene powder reflectance standards. *Applied Optics*, *24*(14), 2225. https://doi.org/10.1364/AO.24.002225 — *PTFE reference repeatability (phase 07)* [N31]
- Hsia, J. J., & Barnes, P. Y. (1995). *45°/0° reflectance factors of pressed polytetrafluoroethylene (PTFE) powder*. https://doi.org/10.6028/NIST.TN.1413 — *45/0 PTFE factor for our geometry (phase 07)* [N32]
- Barnes, P. Y., Early, E. A., & Parr, A. C. (1998b). *Spectral reflectance*. https://doi.org/10.6028/NIST.SP.250-48 — *Traceable reflectance calibration method (phase 07)* [N33]
- Labsphere. (n.d.). *Technical guide: Reflectance materials and coatings*. https://www.systems-eng.co.jp/dcms_media/other/Guide_Reflectance_Materials_Coatings.pdf — *Spectralon vs DIY PTFE choice (phase 07)* [N34]
- Springsteen, A. (1998a). Reflectance spectroscopy: An overview of classification and techniques. In *Applied spectroscopy: A compact reference for practitioners* (pp. 193–224). Academic Press. https://doi.org/10.1016/B978-012764070-9/50008-1 — *Reflectance geometry nomenclature (phase 07)* [N35]

- Springsteen, A. (1998b). Standards for reflectance measurements. In *Applied spectroscopy: A compact reference for practitioners* (pp. 247–267). Academic Press. https://doi.org/10.1016/B978-012764070-9/50010-X — *Choosing white/grey/black references (phase 07)* [N36]

Multispectral imaging with LED illumination

- Park, J.-I., Lee, M.-H., Grossberg, M. D., & Nayar, S. K. (2007). Multispectral imaging using multiplexed illumination. *Proceedings of the IEEE International Conference on Computer Vision*, 1–8. https://doi.org/10.1109/ICCV.2007.4409090 — *Multiplexed LED band capture (phase 10)* [N37]
- Shrestha, R., & Hardeberg, J. Y. (2013). Multispectral imaging using LED illumination and an RGB camera. *Color and Imaging Conference*, *21*(1), 8–13. https://doi.org/10.2352/CIC.2013.21.1.art00003 — *LED bank + RGB camera reflectance (phase 10)* [N38]
- Shrestha, R., & Hardeberg, J. Y. (2014a). Evaluation and comparison of multispectral imaging systems. *Color and Imaging Conference*, *22*(1). https://doi.org/10.2352/CIC.2014.22.1.art00018 — *Spectral accuracy metrics (phases 10, 17)* [N39]
- Shrestha, R., & Hardeberg, J. Y. (2014b). How are LED illumination based multispectral imaging systems influenced by different factors? *ICISP 2014*, *61*, 61–71. https://doi.org/10.1007/978-3-319-07998-1_8 — *Noise/band-count sensitivity of LED bank (phase 10)* [N40]
- Imai, F. H., Berns, R. S., & Tzeng, D.-Y. (2000). A comparative analysis of spectral reflectance estimated in various spaces using a trichromatic camera system. *Journal of Imaging Science and Technology*, *44*(4), 280–287. https://doi.org/10.2352/J.ImagingSci.Technol.2000.44.4.art00003 — *Spectral reconstruction from camera signals (phase 10)* [N41]
- Imai, F. H., Rosen, M. R., & Berns, R. S. (2000). Comparison of spectrally narrow-band capture versus wide-band with a priori sample analysis for spectral reflectance estimation. *Color and Imaging Conference*, *8*(1), 234–241. https://doi.org/10.2352/CIC.2000.8.1.art00043 — *Narrow-band LEDs vs filters trade-off (phase 10)* [N42]
- Hardeberg, J. Y. (2001). *Acquisition and reproduction of color images: Colorimetric and multispectral approaches*. https://doi.org/10.1002/col.10073 — *Camera colorimetric/spectral calibration (phase 10)* [N43]
- Hardeberg, J. Y. (2004). Filter selection for multispectral color image acquisition. *Journal of Imaging Science and Technology*, *48*(2). https://doi.org/10.2352/J.ImagingSci.Technol.2004.48.2.art00005 — *Choosing the 8 LED bands (phase 10)* [N44]

Optics textbooks

- Kingslake, R., & Johnson, R. B. (2010). *Lens design fundamentals* (2nd ed.). https://doi.org/10.1016/B978-0-12-374301-5.00008-5 — *M12 lens f-number/aberration reasoning (phase 06)* [N45]
- Goodman, J. W. (2017). *Introduction to Fourier optics* (1st ed.). https://doi.org/10.1063/1.3035549 — *Diffraction limit and MTF at f/8 (phase 06)* [N46]

Photobiological and machine/electrical safety

- IEC. (n.d.). *Photobiological safety of lamps and lamp systems*. https://doi.org/10.3403/30149289 — *COB and LED-bank eye-safety risk groups (phases 04, 10)* [N47]
- *ISO, "safety of machinery*. (n.d.-a). https://doi.org/10.3403/30265059 — *E-stop category and reset behaviour (phases 01, 02)* [N48]
- *IEC, "safety of machinery*. (n.d.). https://doi.org/10.3403/30238348 — *Power wiring, colours, stop circuit (phase 01)* [N49]
- *ISO, "safety of machinery*. (n.d.-b). https://doi.org/10.3403/30201665 — *Pinch-point/hazard risk assessment (phases 00, 16)* [N50]
- *IEC, "audio/video, information and communication technology equipment*. (n.d.). https://doi.org/10.3403/3 — *Bought-in PSU safety listing (phase 01)* [N51]

Appendix E · Motion control, electronics, open hardware

Background papers and sources, by topic. Format: APA reference — *what this project uses it for*. Full list: Bibliography.

Stepper drivers and microcontrollers

- Analog Devices (Trinamic). (n.d.-b). *TMC2209 datasheet*. https://www.analog.com/media/en/technical-documentation/data-sheets/tmc2209_datasheet_rev1.09.pdf — *Driver registers, UART, current (phase 02)* [H1]
- Analog Devices (Trinamic). (n.d.-a). *AN-002: Parameterization of StallGuard2 & CoolStep*. https://www.analog.com/en/resources/app-notes/an-002.html — *Stall-detection thresholds, crash detection (phases 02, 04)* [H2]
- Dwersteg (Analog Devices/Trinamic), B. (n.d.). *AN-026: Programmable adaptive microstep table*. https://www.analog.com/en/resources/app-notes/an-026.html — *Microstep linearity, even step spacing (phases 08, 09)* [H3]
- Marlin Firmware. (n.d.). *Trinamic drivers*. https://marlinfw.org/docs/hardware/tmc_drivers.html — *StealthChop vs SpreadCycle choice (phase 02)* [H4]
- Espressif Systems. (n.d.-b). *ESP32 technical reference manual*. https://documentation.espressif.com/esp3 — *E4 MCU peripherals, step timing (phase 02)* [H5]
- Espressif Systems. (n.d.-a). *ESP32 series datasheet*. https://www.espressif.com/sites/default/files/docume — *Strapping pins, electrical limits (phase 02)* [H6]
- Raspberry Pi Ltd. (n.d.-b). *RP2040 datasheet*. https://datasheets.raspberrypi.com/rp2040/rp2040-datasheet.pdf — *Pico GPIO, pull-ups, PIO (phase 02)* [H7]
- Raspberry Pi Ltd. (n.d.-a). *Raspberry Pi Pico-series Python SDK: A MicroPython environment for Raspberry Pi microcontrollers*. https://datasheets.raspberrypi.com/pico/raspberry-pi-pico-python-sdk.pdf — *MicroPython button-hub firmware (phase 02)* [H8]

Interface electronics and position sensors

- Sharp Corporation. (2018). *PC817XxNSZ1B series*. https://global.sharp/products/device/lineup/data/pdf/ — *24 V sense optocoupler, CTR (phase 01)* [H9]
- Texas Instruments. (2025). *ULN200x, ULQ200x high-voltage, high-current darlington transistor arrays*. https://www.ti.com/lit/gpn/ULN2003A — *8-band LED low-side switching (phase 10)* [H10]
- Vishay Semiconductors. (n.d.). *TCST2103, TCST2202, TCST2300: Transmissive optical sensor with phototransistor output*. https://www.vishay.com/docs/81147/tcst2103.pdf — *Optical homing flag, 1 mm aperture (phases 04, 05)* [H11]
- ams AG (ams OSRAM). (2014). *AS5600 – 12-bit programmable contactless potentiometer*. https://www.mouser.com/pdfdocs/AMS_AS5600_Datasheet_EN.PDF — *Absolute-angle check of axes (phase 17)* [H12]

Camera interface

- Sony Semiconductor Solutions. (2018). *IMX477-AACK: Diagonal 7.857 mm (type 1/2.3) 12.3 mega-pixel CMOS image sensor*. https://www.sony-semicon.com/files/62/pdf/p-13_IMX477-AACK_Flyer.pdf — *Pixel pitch 1.55 µm, sensor size (phase 06)* [H13]
- USB Implementers Forum. (2012). *Video class v1.5 document set*. https://www.usb.org/document-library/video-class-v15-document-set — *UVC exposure/white-balance controls (phase 06)* [H14]
- Linux kernel documentation. (n.d.). *The linux USB video class (UVC) driver*. https://www.kernel.org/doc/ht api/media/drivers/uvcvideo.html — *uvcvideo controls, extension units (phases 06, 14)* [H15]

Belt drives and positioning accuracy

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